

Modular and Compositional Adaptation of a Multilingual LM

Take-home assignment • PhD student position

Alejandro Pardo-Bascuñana

1 Introduction

Adapting language models to low-resource domains is difficult because specialized text typically combines multiple factors, such as language, time period, and genre (Gururangan et al., 2020). For example, a Spanish legal corpus contains both linguistic regularities in Spanish and aspects of the legal-administrative genre. Standard domain adaptation learns these properties jointly on the target data. While straightforward, this requires training data for each specific intersection and does not allow the reuse of modules learned on other corpora.

We test whether these factors can instead be learned independently and later combined in parameter space. Using the OPUS collection, we chose Spanish as the language factor and legal/administrative text as the genre factor. We independently train:

1. A Spanish module on literary text from OPUS Books (Tiedemann, 2012).
2. A legal module on non-Spanish text from OPUS DGT (Steinberger et al., 2012).

Neither module observes Spanish legal text during training. We then combine them using parameter arithmetic (Ilharco et al., 2023), testing both zero-shot composition (without target data) and lightweight target-domain refinement.

This setup lets us test whether factor composition is viable compared to direct adaptation. A composed model is useful only if it improves upon individual factor modules and remains competitive with a model trained directly on the target intersection. We therefore evaluate the unadapted backbone, both single-factor modules, direct target adaptation, zero-shot composition, and supervised composition.

Our experiments demonstrate two main findings. First, neither factor module improves on the pretrained backbone when evaluated on Spanish legal text; the Spanish literary module causes a clear degradation, while the multilingual legal module performs slightly worse than the base model. Second, direct target adaptation produces the lowest perplexity. Supervised refinement recovers most of the zero-shot composition loss, but remains slightly worse than direct adaptation. These results show that independently learned modules do not cleanly compose to the target intersection in this setup.

2 Experimental Design

2.1 Target domain and corpora

We select the Spanish portion of OPUS DGT as the target intersection because it pairs a widely spoken language

with a specialized administrative genre. We use fixed OPUS language pairs (**es-ga** for Spanish DGT and **en-es** for Spanish Books) to ensure that data splits remain identical across runs.

The target corpus is split deterministically by hashing normalized Spanish texts into 80% training, 10% validation, and 10% test sets. Duplicate texts are removed before splitting, and the sets are verified to be disjoint. Following the assignment setup, the training split contains exactly 1,000,000 whitespace-separated words, while validation and test splits contain 100,000 words each.

The language module uses 1,000,000 Spanish words from OPUS Books. For the genre module, we sample 1,000,000 words balanced across 9 non-Spanish DGT languages (Bulgarian, Finnish, Irish, Croatian, Maltese, Dutch, Serbo-Croatian, Slovak, and Swedish), explicitly excluding Spanish to prevent data leakage.

Table 1 summarizes the resulting datasets. Because tokenizer efficiency varies across languages, 1M words yield 5,824 packed 512-token blocks for legal text, but only 3,263 blocks for Spanish literary text. To ensure equal token exposure and optimizer updates, we shuffle and downsample the training blocks to 3,263 for each factor (and 327 blocks for validation).

2.2 Backbone and adaptation modules

We use Qwen/Qwen3-0.6B-Base¹ (Yang et al., 2025), a pretrained multilingual causal language model with approximately 0.6B parameters. We chose Qwen because models in this family achieve competitive performance on multilingual benchmarks, including the Massive Text Embedding Benchmark (MTEB) leaderboard (Muennighoff et al., 2023)². Although many top-ranked MTEB models are non-generative embedding encoders, our language modeling objective requires an autoregressive generative model. Given our compute constraints on Kaggle, Qwen3-0.6B provides the necessary balance between strong multilingual representations and lightweight training efficiency.

We implement each module using Weight-Decomposed Low-Rank Adaptation (DoRA) (Liu et al., 2024). DoRA separates the magnitude and directional components of pretrained weights and applies a low-rank update (Hu et al., 2022) to the direction, allowing the modules to remain parameter-efficient during training while letting us merge updates into complete weight deltas for composition.

¹<https://huggingface.co/Qwen/Qwen3-0.6B-Base>

²<https://huggingface.co/spaces/mteb/leaderboard>

Corpus	Split	Documents	Words	Available	Used
Spanish OPUS Books	Train	37,317	1,000,000	3,263	3,263
Spanish OPUS Books	Validation	4,663	100,000	334	327
Non-Spanish OPUS DGT	Train	46,827	1,000,000	5,824	3,263
Non-Spanish OPUS DGT	Validation	4,861	100,000	583	327
Spanish OPUS DGT	Train	32,466	1,000,000	3,272	3,272
Spanish OPUS DGT	Validation	3,296	100,000	327	327
Spanish OPUS DGT	Test	3,228	100,000	326	326

Table 1: Corpora retained after normalization and word-budget sampling. “Available” gives the number of complete 512-token blocks after packing; “Used” gives the number retained for training, validation, or testing.

Using Hugging Face PEFT (Mangrulkar et al., 2022), we attach DoRA adapters to (`all-linear`) layers with rank $r = 8$, scaling factor $\alpha = 16$, dropout 0.05, and no trainable bias. In total, DoRA wraps 196 linear layers and introduces 5,390,336 trainable parameters (0.8962% of the 601M base parameters).

Texts are tokenized and concatenated with end-of-sequence tokens into non-overlapping 512-token blocks. We train for up to three epochs using AdamW (Loshchilov and Hutter, 2019) with an effective batch size of 64 (batch size 1 per GPU with 32 accumulation steps), a learning rate of 2×10^{-4} , cosine decay, weight decay of 0.01, and gradient clipping at 1.0. Validation is checked twice per epoch (every 26 optimizer steps). Validation loss improved consistently across all three epochs without triggering early stopping, resulting in 153 updates for the factor modules and 156 for the target modules.

2.3 Independent training and composition

The Spanish and legal modules are trained independently from identical copies of the backbone W_0 . Let W_{es} and W_{legal} denote the weights obtained after merging their respective DoRA modules. To combine both capabilities without target training data, we apply task arithmetic (Ilharco et al., 2023) by summing their parameter deltas:

$$\begin{aligned} W_{\text{comp}} &= W_0 + (W_{\text{es}} - W_0) + (W_{\text{legal}} - W_0) \\ &= W_{\text{es}} + W_{\text{legal}} - W_0. \end{aligned} \quad (1)$$

All weight arithmetic is performed in float32 on CPU to prevent precision loss.

For supervised composition, we freeze the zero-shot merged weights W_{comp} and attach a new DoRA module ($r = 8$) trained on the 1M-word Spanish DGT training split. We compare this against direct target adaptation, in which an equivalent DoRA module is trained on the same target data, starting from W_0 . This checks whether the composed initialization provides any benefit over standard adaptation.

2.4 Baselines and evaluation

We evaluate six conditions: the unadapted backbone, Spanish DoRA, legal DoRA, direct Spanish-legal DoRA, zero-shot composition, and composition followed by supervised DoRA. All models are evaluated on the same 326 held-out Spanish DGT test blocks.

For each block, we compute the mean next-token Negative Log-Likelihood (NLL) and report Perplexity ($\text{PPL} = \exp(\text{mean NLL})$). We estimate 95% confidence intervals with 5,000 bootstrap resamples across test blocks. We also perform paired bootstrap tests with 10,000 resamples over block-level NLL differences ($\Delta\text{NLL} = \text{NLL}_A - \text{NLL}_B$), which accounts for variation in block difficulty. A negative difference favors Model A.

To verify that the factor adapters trained successfully, we also evaluate each module on its respective source validation split (Spanish Books for Spanish DoRA and non-Spanish DGT for Legal DoRA, each using 327 blocks).

3 Results

Table 2 presents the evaluation on the Spanish DGT test set, and Figure 1 visualizes the comparison on a log scale. Only the target-supervised conditions improve over the base model. The individual factor modules and their zero-shot composition all result in higher perplexity on the target domain.

Table 3 reports the paired block-level comparisons. All confidence intervals exclude zero and all differences are positive, confirming that Model A performs worse than Model B in each comparison. Figure 2 highlights the two key composition comparisons: zero-shot composition is clearly worse than Legal DoRA alone, while supervised composition is slightly worse than direct target adaptation.

3.1 Source-domain adaptation

Table 4 verifies that both factor modules adapt successfully to their training domains. Spanish DoRA reduces perplexity on Spanish Books from 30.69 to 25.65 (a 16.4% improvement; paired $\Delta\text{NLL} = -0.1794$).

Legal DoRA produces a larger in-domain improvement, reducing non-Spanish DGT perplexity from 28.41 to 14.63 (a 48.5% improvement; paired $\Delta\text{NLL} = -0.6637$). Both paired bootstrap tests yield $p = 0.0002$, confirming that the target-domain failures are not caused by flawed factor training.

Model	Mean NLL	Perplexity	95% bootstrap CI	Reduction vs. base
Base	2.5462	12.76	[12.46, 13.05]	–
Spanish DoRA	2.8371	17.07	[16.60, 17.53]	–33.8%
Legal DoRA	2.5572	12.90	[12.61, 13.18]	–1.1%
Zero-shot composition	2.7047	14.95	[14.58, 15.31]	–17.2%
Direct Spanish-legal DoRA	2.0983	8.15	[7.94, 8.36]	36.1%
Composition + supervised DoRA	2.1096	8.24	[8.03, 8.45]	35.4%

Table 2: Performance on 326 held-out Spanish DGT blocks. Lower NLL and perplexity are better. Negative reductions indicate degradation relative to the base model. Confidence intervals are obtained by bootstrapping block losses.

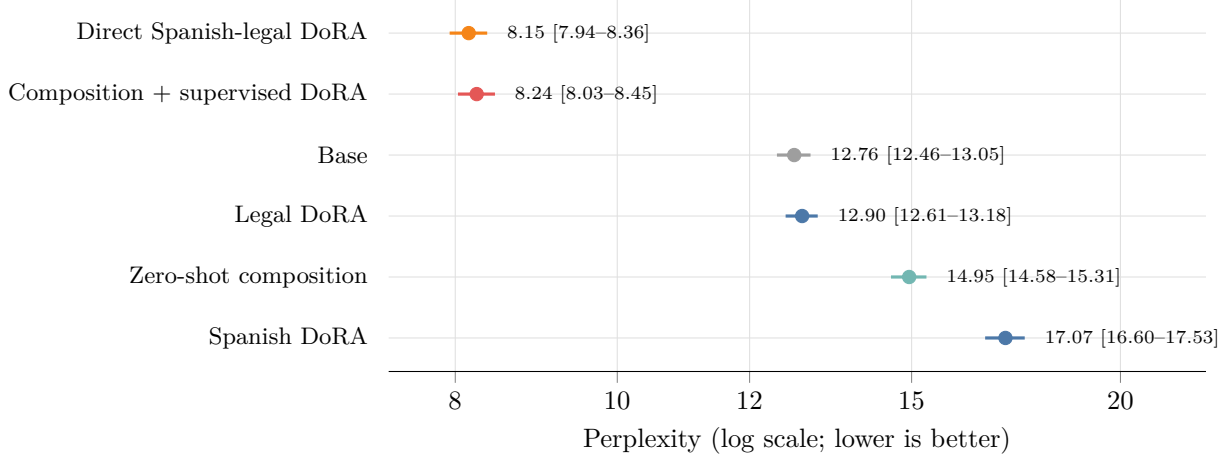


Figure 1: Perplexity on held-out Spanish DGT. Points denote mean perplexity and horizontal lines denote 95% bootstrap confidence intervals.

Model A	Model B	$\Delta\text{NLL} (A - B)$	95% paired CI	p
Spanish DoRA	Base	+0.2909	[+0.2845, +0.2974]	0.0002
Legal DoRA	Base	+0.0110	[+0.0050, +0.0173]	0.0006
Zero-shot composition	Legal DoRA	+0.1475	[+0.1423, +0.1526]	0.0002
Composition + supervised DoRA	Direct Spanish-legal DoRA	+0.0113	[+0.0098, +0.0127]	0.0002

Table 3: Paired bootstrap comparisons over the same 326 test blocks, using 10,000 resamples. Positive differences favor Model B. The minimum attainable two-sided p -value with the applied pseudocount is approximately 0.0002.

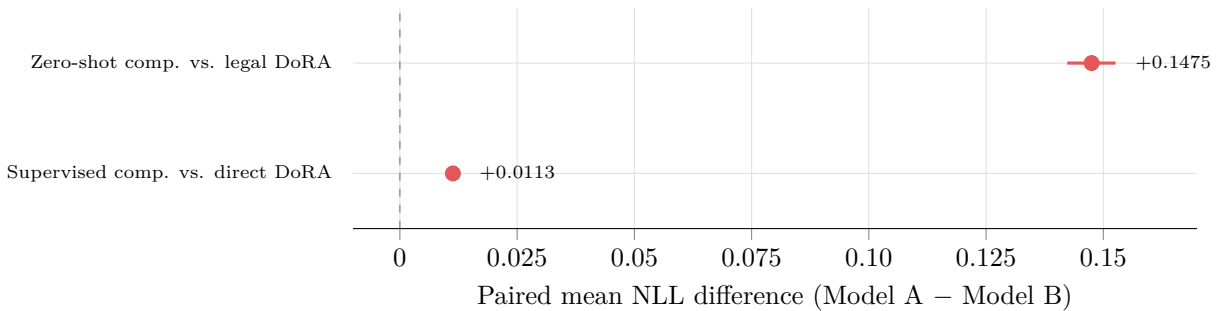


Figure 2: Paired NLL differences for the two composition comparisons. Horizontal lines denote 95% paired-bootstrap confidence intervals; positive values favor the second model named in each row.

3.2 Factor-specific adaptation

Although Spanish DoRA improves in-domain, it degrades target performance on Spanish DGT, increasing perplexity from 12.76 to 17.07 (+33.8%; paired $\Delta\text{NLL} = +0.2909$). This indicates that adapting to literary text introduces stylistic features that conflict with administrative language.

Legal DoRA reduces non-Spanish DGT perplexity by 48.5%, but achieves 12.90 on Spanish DGT (+1.1% vs. base; paired $\Delta\text{NLL} = +0.0110$). While the difference is small, it shows that multilingual legal knowledge does not transfer directly to Spanish legal text without Spanish-specific training.

Table 4: Source-domain evaluation. Perplexity intervals and paired NLL intervals are obtained by bootstrap resampling of validation blocks. Paired differences are calculated as adapter minus backbone, so negative values favor the adapter.

Validation domain	Model	Mean NLL	Perplexity	95% PPL CI	Change vs. base
Spanish Books	Base	3.4240	30.69	[30.22, 31.19]	–
Spanish Books	Spanish DoRA	3.2446	25.65	[25.27, 26.04]	–16.4%
Non-Spanish DGT	Base	3.3467	28.41	[25.83, 31.29]	–
Non-Spanish DGT	Legal DoRA	2.6829	14.63	[13.81, 15.50]	–48.5%

Validation domain	Adapter	Paired Δ NLL	95% paired CI	p -value
Spanish Books	Spanish DoRA	–0.1794	[–0.1881, –0.1710]	0.0002
Non-Spanish DGT	Legal DoRA	–0.6637	[–0.7175, –0.6124]	0.0002

3.3 Zero-shot composition

Zero-shot composition yields a perplexity of 14.95. While better than Spanish DoRA alone (17.07), it is 15.9% worse than Legal DoRA (12.90) and 17.2% worse than the base model (12.76).

3.4 Target-supervised adaptation

Direct Spanish-legal DoRA achieves the best perplexity at 8.15 (a 36.1% reduction vs. base). This substantial gain confirms that DoRA adapts effectively when provided with matched target data.

Supervised refinement of the composed model reaches 8.24 (a 35.4% reduction vs. base). While the margin is small, direct adaptation consistently outperforms composition refinement across all validation checkpoints and test blocks (paired Δ NLL = +0.0113, $p = 0.0002$). Initializing from the composed weights, therefore, provides no advantage over standard adaptation from the backbone.

Figure 3 displays the loss curves for all four training runs. All runs exhibit steady convergence and stable validation loss without overfitting.

4 Discussion

Our results evaluate two forms of compositional adaptation, neither of which improves over the unadapted backbone. Zero-shot parameter addition is clearly worse than Legal DoRA alone and worse than the base model. Target supervision repairs most of this degradation, but refining from composed weights remains slightly worse than training DoRA directly from the base model.

The source-domain checks show that the issue is not optimization failure: both factor modules improve on their training distributions (reducing perplexity by 16.4% and 48.5%). Rather, the adapters capture patterns specific to their training domains. The Spanish update captures literary style that conflicts with legal text, while the legal update captures multilingual administrative patterns that do not transfer zero-shot to Spanish without target grounding.

Some limitations apply to our study. We evaluate one language-genre pair and one random seed. In addition, perplexity assesses the fit of next-token probabilities, but

it does not guarantee that the model can actually answer legal questions, summarize a case, or avoid hallucinations.

Future work could explore learned merging coefficients (e.g., TIES-merging; [Yadav et al., 2023](#)), layer-wise delta analysis to identify parameter conflicts (measuring changes layer-by-layer), and evaluating how well the models perform on specific legal applications, such as classification or summarization.

5 Conclusion

We independently adapted **Qwen3-0.6B-Base** to Spanish literary and non-Spanish legal text, combining them via parameter arithmetic to enable Spanish legal modeling. Although both adapters learned their respective source domains effectively, zero-shot composition failed in the target domain, resulting in higher perplexity than the unadapted base model. Supervised target adaptation achieved strong gains (perplexity 8.15), but initializing from composed weights provided no benefit over direct adaptation. These results show that independent factor adaptation does not guarantee factorized transfer through simple parameter arithmetic.

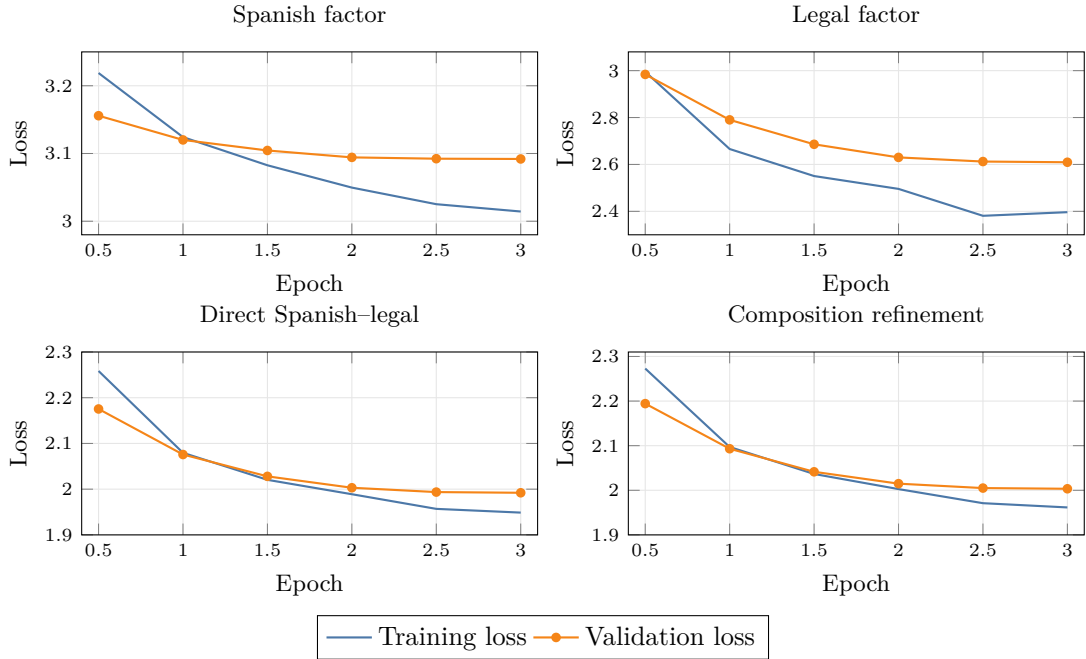


Figure 3: Training and validation loss for the four adapter-training runs. Values are reported at the twice-per-epoch validation schedule. Vertical scales differ across panels.

A Appendix: Backbone Sensitivity with LFM2.5-350M-Base

To evaluate whether our findings generalize across architectures, we replicated the experiment using LiquidAI/LFM2.5-350M-Base³ (LFM). The data splits, token budgets, and DoRA rank configurations ($r = 8$, $\alpha = 16$) remain identical to the main Qwen setup, with training set to a learning rate of 1×10^{-4} and a maximum of 5 epochs.

Table 5: Spanish DGT evaluation across LFM2.5-350M and Qwen3-0.6B backbones (Δ denotes PPL change relative to Base).

Condition	LFM2.5-350M		Qwen3-0.6B	
	PPL	Δ	PPL	Δ
Base	97.22	–	12.76	–
Spanish DoRA	23.93	–75.4%	17.07	+33.8%
Legal DoRA	18.18	–81.3%	12.90	+1.1%
Zero-shot comp.	23.01	–76.3%	14.95	+17.2%
Direct DoRA	8.09	–91.7%	8.15	–36.1%
Comp. + supervised	8.00	–91.8%	8.24	–35.4%

Table 5 compares target performance. Single-factor adapters behave differently across backbones: on LFM, both modules yield massive perplexity reductions (e.g., Legal DoRA drops PPL from 97.22 to 18.18), whereas on Qwen they offer no target benefit. This discrepancy reflects backbone capacity and pretraining coverage; LFM lacks native support for the non-Spanish legal source languages, causing its legal adapter to function partly as general multilingual alignment.

More importantly, zero-shot composition fails on both backbones: unweighted addition is worse than Legal

DoRA alone on LFM (23.01 vs. 18.18, $\Delta\text{NLL} = +0.2354$) and on Qwen (14.95 vs. 12.90, $\Delta\text{NLL} = +0.1475$). Furthermore, supervised refinement yields minor and inconsistent differences relative to direct adaptation (-0.0109 on LFM vs. $+0.0113$ on Qwen). These results confirm that factorized adaptation is sensitive to base model competence and that unweighted parameter arithmetic is not a robust substitute for target-domain supervision.

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Shih-Yang Liu, Chien-Yi Wang, Hongxu Yin, Pavlo

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